Sahaj Bhadja

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Summary

Software engineer with 2+ years building **C++/Python systems**, computer-vision and ML applications, automated test tooling, and cross-platform software; experienced in system integration, performance analysis, root-cause diagnosis, documentation, and CI/CD.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Developed **Python data-processing services and automated analysis workflows** for 500+ users, optimizing indexed retrieval and software performance to reduce latency from 1.2 seconds to 350 milliseconds across production systems.
- Created **validation, reconciliation, and anomaly-diagnosis tooling** across 5,000+ monthly jobs, analyzing logs and failure states to accelerate root-cause analysis while achieving 99%+ reliability through tested CI/CD releases.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Built a **concurrent Python processing system** using multithreading and fault-tolerant APIs, validating PDF, DOCX, and ZIP inputs while handling 1,000+ files per batch and 10,000+ daily requests.
- Integrated **multimodal AI, computer-vision, and OCR tools** behind adaptable APIs, adding usage logging, persistent results, fallback handling, and automated tests to improve analysis quality while reducing turnaround time 60%.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Engineered **Python simulation and evaluation pipelines** processing 100,000+ autonomous-system trajectories, integrating map geometry and temporal signals while developing reproducible test scenarios that improved long-horizon planning performance 18%.
- Analyzed **system constraints, benchmark results, and failure patterns** across three ML architectures, documenting findings and recommending software improvements that strengthened closed-loop stability 15% through quantitative, data-driven performance feedback.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Built **cross-platform object-oriented software components** using API integrations, SQL, caching, Singleton, and Factory patterns, improving response times 30%, pipeline efficiency 68%, storage costs 15%, and maintainability across distributed environments.
- Automated **builds, deployments, and 20+ unit/integration tests** with GitHub, Jenkins, PyTest, and Mockito, generating coverage, CSV reports, and failure logs across Windows, Linux, and macOS production releases.

Cyber Security and Data Mining **India**
Data Analyst - Research Fellowship *May 2023 – August 2023*
- Developed a **high-volume anomaly classification pipeline** over 650,000+ records, engineering 21 statistical and behavioral features and comparing Random Forest, Naive Bayes, RNN, LSTM, GRU, and CNN architectures.
- Achieved **96%+ detection accuracy** through reproducible model evaluation, quantitative performance analysis, and feature-level diagnostics, documenting actionable phishing and malware patterns while continuously refining data-processing and validation methods.

Skills

Languages: C++, C, Python, Java, SQL, Bash, JavaScript, TypeScript
AI/ML & Vision: PyTorch, TensorFlow, Transformers, CNNs, RNNs, computer vision, OCR, feature engineering
Systems: Linux, Windows, OOP, data structures, algorithms, multithreading, concurrency, simulation, performance analysis
Quality: Root-cause analysis, unit testing, integration testing, PyTest, Mockito, Git, Jenkins, CI/CD, documentation
Platform: REST APIs, FastAPI, AWS, Azure, Docker, GitHub Actions, Agile/Scrum, AI-assisted development

Projects

CPU Scheduling Simulator Live | *C++, React.js, OS Internals*
- Implemented and validated **six C++ scheduling algorithms** with interactive controls, Gantt visualization, edge-case handling, and dynamic performance metrics across 1,000+ process combinations, reducing average turnaround time 25% and debugging time 20%.

CaptionCraft Live | *Python, PyTorch, Transformers, Flask*
- Evaluated **four transformer vision architectures** across 10,000 images and 100,000 captions, improving accuracy from 81% to 96% while building Flask inference APIs for scalable real-time caption generation.

PiXMod Live | *PyTorch, SAM, Stable Diffusion, MongoDB*
- Built an **image-processing and segmentation pipeline** using SAM, Stable Diffusion, and PyTorch, processing 1,000+ edited assets while storing images and metadata in MongoDB for scalable retrieval and analysis.